

Author Contribution Statement

Singular Value Interlacing for Physics-Structured Projections: Condition-Number Inheritance and the Physics-Reduction Proposition

CRedit Taxonomy

Contribution	A. V. Eluvathingal	R. K. Prasad
Conceptualisation	Lead	Supporting
Formal analysis	Lead	—
Funding acquisition	—	Lead
Investigation	Lead	—
Methodology	Lead	Supporting
Project administration	—	Lead
Software (numerical expts)	Lead	—
Supervision	—	Lead
Validation	Lead	—
Visualisation	Lead	—
Writing — original draft	Lead	—
Writing — review & editing	Supporting	Lead

Plain-Language Statement (for portal upload)

Arun V. Eluvathingal: Identified the need for SVD interlacing bounds for non-orthonormal physics-structured projections arising from DAE constraint elimination; formulated and proved Theorem 1 (interlacing with coupling correction), Theorem 2 (condition-number inheritance), Corollary 1 (sufficient inheritance condition), and Proposition 1 (Physics-Reduction Proposition for the IBR fault-current Jacobian); proved sharpness of all bounds; carried out all numerical experiments (1000-point Monte Carlo on the Park dq measurement map); produced all figures; wrote the manuscript.

R. K. Prasad: Supervised the research; provided physical intuition connecting the algebraic coupling ratio ρ to generator inductance parameters; reviewed and edited the final manuscript; secured project funding.

Conflict of interest: The authors declare no conflict of interest.

Data and code: MATLAB code for computing the physics-structured projection $\mathbf{P}$, singular values, and condition numbers, together with the 1000-point experiment scripts, will be deposited at <https://zenodo.org> upon acceptance.